



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

Course Code : SECD2613 Section 01
Course Name : System Analysis and Design
Lecturer : Dr Muhammad Aliff
Academic Session : 2025/2026 Semester 2
Assessment Title : Project Phase 1 (FlexRent Group)
Submission Due Date : 20 April 2026

Prepared by

:

Matric No.	Student Name
B22EC0071	Mohamad Azryq Haiqal Bin Mohd Rosli
A25CS0046	Bazli Abdullah Bin Mohamad Yusoff
A25CS0257	Muhamad Afif Azfar Bin Mohd Shahrizan
A25CS0255	Mikhail Adam Riaz Bin Yazmi
A25CS0179	Ahnaf Bin Zaizal

Assessment Mode

:

~~Individual~~ / Group

Feedback from Lecturer:

Table of Contents

1.0	Introduction	3
2.0	Background Study	3
2.1	Organizational Background	3
3.0	Problem Statement.....	4
4.0	Proposed Solution	4
4.1	Technical Feasibility	4
4.2	Operational Feasibility	5
4.3	Economic Feasibility	5
5.0	Objectives.....	7
6.0	Scope of The Project	8
7.0	Project Planning.....	9
7.1	Human Resource.....	9
7.2	Work Breakdown Structure (WBS).....	9
7.3	PERT Chart	10
7.4	Gantt Chart.....	11
8.0	Benefits and Overall Summary of Proposed System	11
Github		12

1.0 Introduction

The digital transformation of service-based industries has significantly accelerated the demand for sophisticated, technology-driven platforms. These platforms are essential for managing increasingly complex transactional processes with both efficiency and adaptability. As the nature of service delivery evolves, there is a growing expectation that technology will not only streamline operations but also respond dynamically to diverse and changing user needs. Consequently, the development and implementation of intelligent systems that can seamlessly integrate various service components have become critical to maintaining competitive advantage and ensuring high-quality customer experiences in this rapidly changing landscape.

Within the domain of property rentals, traditional systems have exhibited notable limitations, particularly in addressing the diverse and constantly evolving scheduling demands of modern users. These platforms often lack the flexibility required to accommodate the complex and dynamic nature of current rental arrangements, thereby restricting their effectiveness in meeting user expectations for adaptability and convenience. These legacy platforms often operate on rigid, predefined booking structures that fail to address the specific and changing time-related (temporal) and situational (contextual) requirements of both service providers and service renters. The operations need to rely entirely on a manual third-party messaging and cross-referencing.

FlexRent has been introduced as a property rental system designed with AI features and limitless flexibility bookings for users based on hourly, daily and monthly durations. The system enables users to function as both property seekers and providers, facilitating a seamless and intelligent rental experience. It achieves this by leveraging AI-driven recommendations that use user-provided keywords to identify and address their specific needs accurately. Moreover, we included an AI algorithm that recommends users based on past bookings and searches.

2.0 Background Study

2.1 Organizational Background

The proliferation of on-demand digital marketplaces in recent years has fundamentally reshaped consumer expectations regarding the accessibility, convenience, and personalisation of rental and service-oriented platforms. Current property rental services such as Airbnb, PropertyGuru and Booking.com only focus on specific rental durations such as daily or monthly. This has the flaw of lacking flexibility for users who want to rent hourly, daily, or monthly in one place. Users seeking to rent a venue for a few hours while simultaneously requiring the services of a professional consultant or event specialist are compelled to navigate multiple disconnected platforms, resulting in operational inefficiency and a fragmented user experience.

FlexRent exists to break these limitations to bring all in one place while providing AI-based recommendations for users to use and AI time slot and management for providers to manage their property more easily and efficiently. Despite this growth, locally developed platforms have yet to fully leverage AI capabilities to deliver adaptive, intelligent rental experiences that respond dynamically to user behaviour and preference.

3.0 Problem Statement

Despite advancements in digital marketplace platforms, existing systems remain limited in flexibility, personalization, and efficiency. Users encounter several issues, including:

- Users struggle to find suitable options efficiently due to limited personalization
- Users have to manually cross-reference their physical notebooks or personal digital calendars before replying.
- Users have to perform an online transfer and make manual transactions via WhatsApp.
- Booking conflicts occur frequently due to poor time-slot management.
- Manual booking modifications, expecting custom penalty rules, collect the user's details and execute a return transfer for different rental times, such as daily and monthly.
- Dependence on multiple platforms to fulfil different requirements.

These limitations result in reduced user satisfaction and inefficient resource utilization. Therefore, there is a need for an AI-integrated platform that can provide flexible booking options, intelligent recommendations, and efficient scheduling. Furthermore, FlexRent provides an all-in-one solution for different rental periods.

4.0 Proposed Solution

The proposed solution is the development of FlexRent, which integrates advanced features to address the identified problems.

4.1 Technical Feasibility

The system can be developed using widely available and reliable technologies:

- Frontend: Python
- Backend: Streamlit
- Database: SQLite
- AI Techniques: Rule-based algorithm

The system will be developed using Python with the Streamlit framework, which allows rapid development of web-based applications without requiring separate frontend and backend implementations. SQLite will be used as the database due to its simplicity and ease of integration. AI functionalities will be implemented using rule-based algorithms to provide basic recommendations. This approach ensures faster development, reduced complexity, and suitability for project implementation.

4.2 Operational Feasibility

The system is designed to be user-friendly and efficient:

- Provides an intuitive interface for both seekers and providers
- Reduces manual effort through automation
- Improves booking efficiency with smart scheduling
- Enhances user satisfaction through personalized recommendations

Thus, the system is practical and can be effectively implemented in real-world environments.

4.3 Economic Feasibility

The benefits significantly outweigh the costs, making the project economically reliable for the system development:

- Cost-Benefit Analysis (CBA)

Assumptions	
Discount rate	10%
Annual increment (cost)	10%
Annual increment (benefit)	4%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefit)	0.9
Estimated cost	
Development	
Hardware	15000
Software	30000
Testing and deployment	5000
Production	
Server and cloud hosting	3600
Domain and security	400

Marketing and ads	12000
Maintenance	6000
Estimated benefits	
Commission fee	60000
Listing fees	5000
Penalty fees	3000
Advertising fees	10000

A. Development cost	Year						
	0	1	2	3	4	5	6
Hardware	16500	-	-	-	-	-	-
Software	33000	-	-	-	-	-	-
Testing and deployment	5500	-	-	-	-	-	-
Total	55000	-	-	-	-	-	-
B. Production cost	Year						
	0	1	2	3	4	5	6
Server and cloud hosting	-	3960	4356	4792	5272	5800	6380
Domain and security	-	440	484	533	587	646	711
Marketing and ads	-	13200	14520	15972	17570	19327	21260
Maintenance	-	6600	7260	7986	8785	9664	10631
Total	-	24200	26620	29283	32214	35437	38982

Present value (PV)	-	22000	22000	22001	22003	22004	22005
Accumulated cost	-	77000	99000	121001	143004	165008	187013
C. Benefits	Year						
	0	1	2	3	4	5	6
Commission fee	-	54000	56160	58407	60744	63174	65701
Listing fees	-	4500	4680	4868	5063	5266	5477
Penalty fees	-	2700	2808	2921	3038	3160	3287
Advertising fees	-	9000	9360	9735	10125	10530	10952
Total	-	70200	73008	75931	78970	82130	85417
Present value (PV)	-	63819	60338	57049	53938	50997	48216
Accumulated benefits	-	63819	124157	181206	235144	286141	334357
Gain/Loss		-13181	25157	60205	92140	121133	147344
Probability Index (PI)	2.68						

5.0 Objectives

The objectives of the proposed systems are:

- To develop a flexible property rental system supporting hourly, daily, and monthly usage.
- To integrate AI for personalized recommendations and smart scheduling.
- To reduce booking conflicts through efficient time management
- To create a unified platform for property rentals
- To enhance overall system usability and efficiency

6.0 Scope of The Project

Based on the discussions conducted, certain features or system components will be developed within the project scope, while others will be identified as falling outside the project coverage.

In Scope

The following are included in the project:

Users

The system will serve the following user groups:

- Renters
- Property provider

Platform

The main deliverable of the project will be a web-based application, with the possibility of developing a mobile application as a future extension.

Key Features

- Smart Schedulers
- Smart Matching
- Flexible schedule system (hourly, daily, monthly and yearly rental)
- Geospatial Proximity Search System
- Property analytics and comparison system
- Rent Budget Comparison system
- Resource Booking System (RBS)

Out of Scope

The following components are excluded from this project:

- Physical properties maintenance - Facilities maintenance and repair of the rental unit are not within the current project scope.
- Furniture rental - It is a separate supply chain and inventory business model that sits apart from the digital booking or matching process.
- Conflict management - These fall outside our software's scope, as it requires legal expertise and compliance with local regulations.
- Utility management & billing - this involves external utility companies and complex billing cycles that are often kept separate from the base rent or booking fee.

7.0 Project Planning

7.1 Human Resource

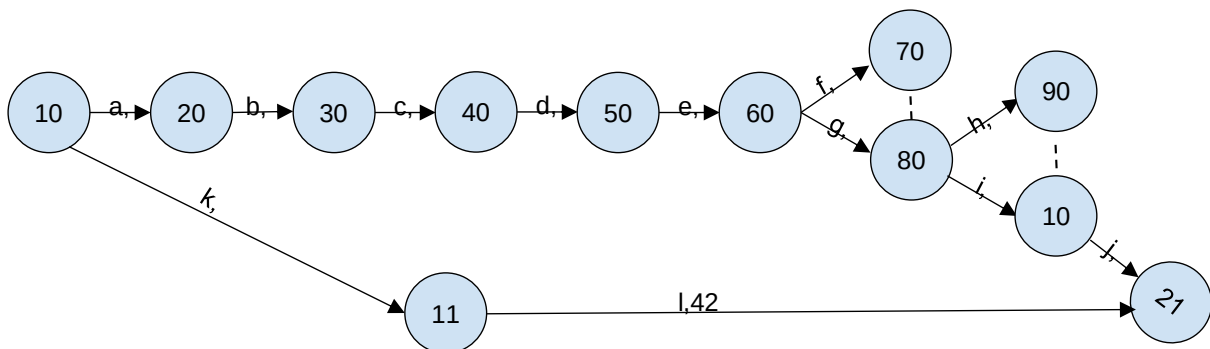
Role	Responsibility	Primary Deliverables
Project Manager	Monitors the project timeline, manages the GitHub Kanban board, and ensures team communication and task completion.	Gantt Chart, PERT Chart, GitHub Management.
System Analyst	Conducts background research and defines the system requirements, specifically the "buffer time" logic for hourly rentals.	Problem Statement, Objectives, Requirement Specs.
AI Specialist	Designs the logic for the AI features and the data flow between system components.	Data Flow Diagrams (DFD), AI Logic Flow.
UI/UX Designer	Designs the user interface for both Seekers and Providers, ensuring a seamless booking experience.	High-Fidelity Figma Prototype.
AI Specialist	Designs the structure for storing property data, user roles, and AI metadata.	Entity Relationship Diagram (ERD).

7.2 Work Breakdown Structure (WBS)

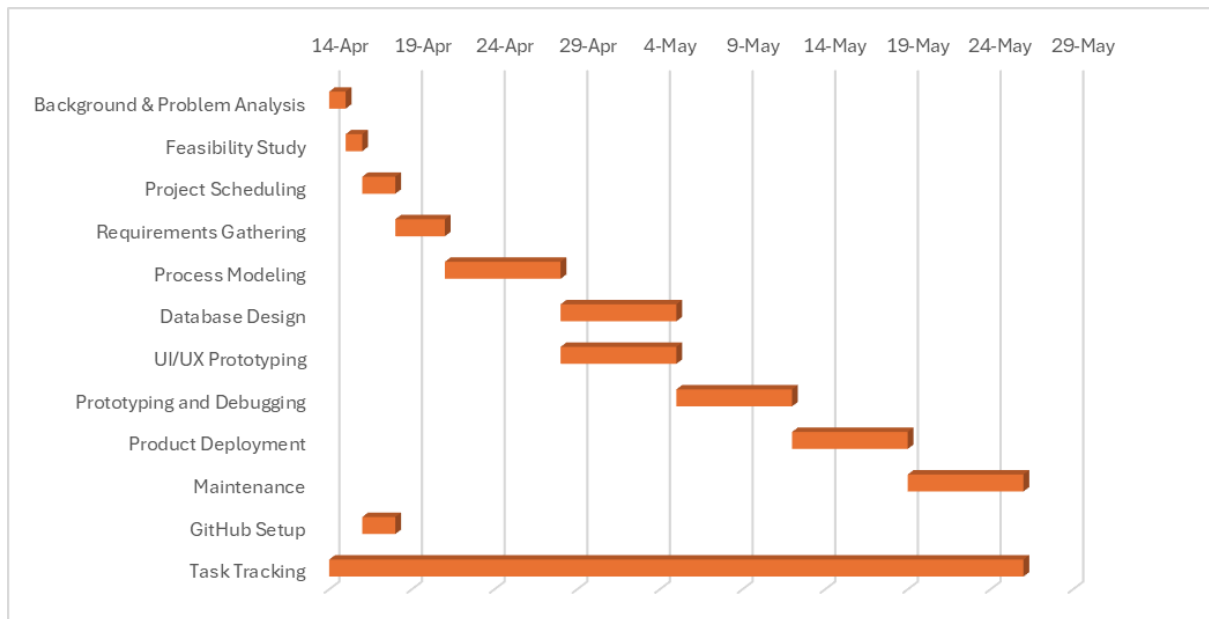
Phase ID	Phase	Activity	Predecessor	Duration (Days)	Date
a	Background & Problem Analysis	Conduct organizational study and define the problem statement.		1	14/4
b	Feasibility Study	Perform Technical, Operational, and Economic (CBA) analysis.	a	1	15/4
c	Project Scheduling	Create WBS, PERT, and Gantt charts.	b	2	16/4
d	Requirements Gathering	Define Functional and Non-Functional requirements.	a, b, c	3	18/4
e	Process Modeling	Create Context Level and Level 0 DFDs showing AI and	d	7	21/4

		booking logic.			
f	Database Design	Create an ERD supporting units of measure (Hour/Day/Month) and AI metadata.	e	7	28/4
g	UI/UX Prototyping	Build a high-fidelity, interactive Figma prototype for Seekers and Providers.	f	7	28/4
h	Prototyping and debugging	Conduct tests on prototype to ensure no errors occur before the final product	f,g	7	5/5
i	Product deployment	Deploy finalised product to production server	g	7	12/5
j	Maintenance	Update system regularly based on user complaints	i	7	19/5
k	GitHub Setup	Initialize repository and invite team/lecturer.		2	16/4
l	Task Tracking:	Manage the Kanban board and perform weekly progress updates.	k	42	14/4

7.3 PERT Chart



7.4 Gantt Chart



8.0 Benefits and Overall Summary of Proposed System

FlexRent presents a comprehensive and innovative solution to the limitations of existing property rental platforms. By integrating AI capabilities with flexible booking systems, it enhances efficiency, reduces operational challenges, and improves user satisfaction.

The system is technically feasible, operationally practical, and economically beneficial, making it a strong candidate for implementation in real-world scenarios. Its scalable design also allows for future enhancements, ensuring long-term relevance and adaptability.

Github

Github project link: https://github.com/AhnafZaizal/Nautica_Project1_SAD_20252026

The screenshot displays the GitHub interface for the repository 'Nautica_Project1_SAD_20252026'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main'. The README file is the selected file, showing a commit by AhnafZaizal 6 minutes ago. The README content includes the project title 'Nautica_Project1_SAD_20252026', a subtitle 'SECD2613 - System Analysis and Design', and a 'Team Members' section with a table listing five team members and their roles.

Team Members	Roles
Mohamad Azryq Haiqal Bin Mohd Rosli	Project Manager
Mikhail Adam Riaz Bin Yazmi	System Analyst
Bazli Abdullah bin Mohamad Yusoff	AI Specialist
Ahnaf Bin Zaizal	UI/UX Designer
Muhamad Afif Azfar bin Mohd Shahrizan	Database Designer